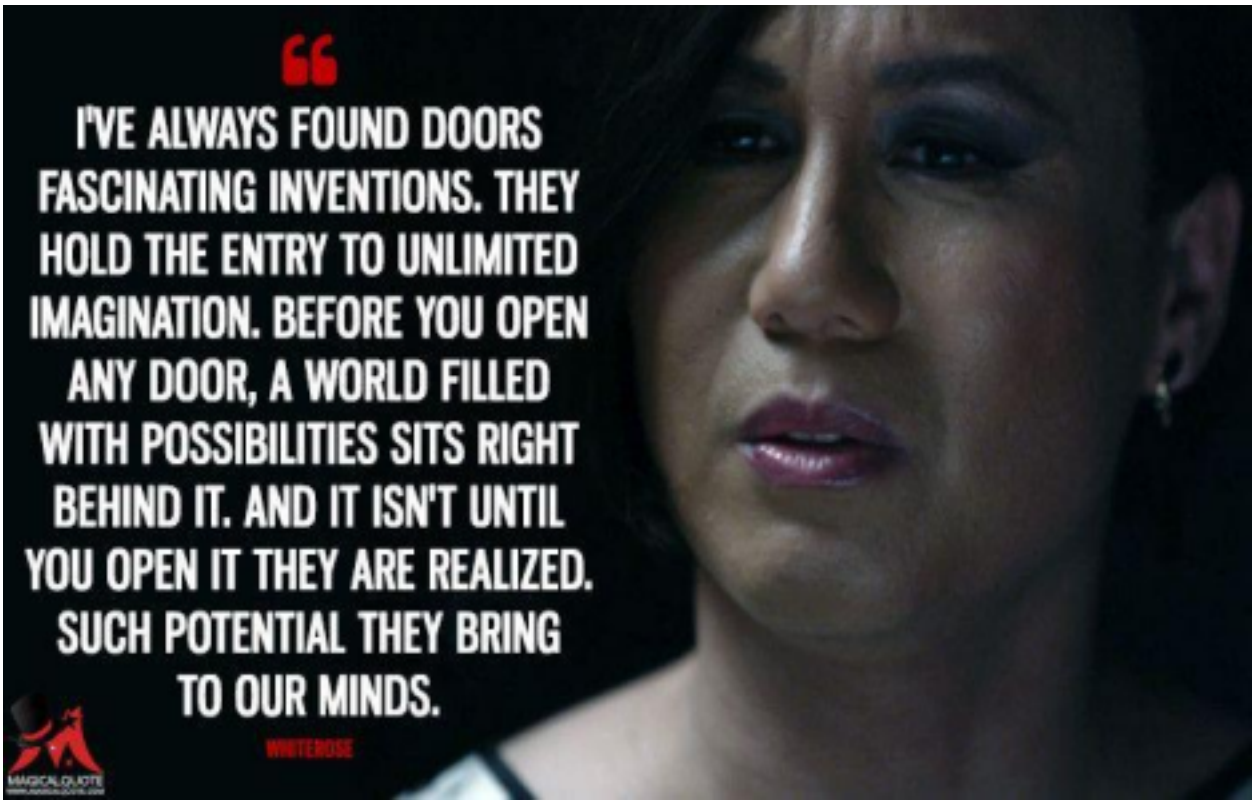
How To ACE TU Ilmeanu RCSE interview



Author: White Rose : Mohammed Ali Java IF YOU ARE AT THE TOP YOU SHOULD NEVER STOP

Big Thanks to FB group: **Study & Life in Germany (For Pakistani Students and Community)**

Introduction:

I have to get full points in this interview and by following these steps I managed to get it. I hope this will help you. This community gave me a lot and I would like to give back. And here is how to ace your RCSE Tu ilmenau interview. Let's get started.

Topics:

S/W engineering Waterfall model - advantages and disadvantages UML – which diagram represents dynamic flow of data. Database

Questions:

- Normalization Algorithms - Sorting techniques - Quick sort - Complexity Networks - Difference b/w layer2 and layer4 - Wireless protocols

1. Data Structure and Algorithms, Complexity. 2. Database Management. 3. Computer Networks. 4. Operating Systems. 5. Software Engineering and UML

Story Line:

Q1:

He started the interview by asking me about what is my motivation for joining this course, where do I see myself after the course. Then he asked me about the technical part.

Note:

Prof Li's specialty is Machine learning so tell him I wanted Deep learning in this program that's what has attracted me the most. Tell him what he wants to hear

Q2:

What is the difference between space and time complexity ? What is binary search ? What is the complexity of binary search?
Answer:

O notation best time scenario

Omega the worst case scenario

Theta which is an average scenario.

Binary search what the complexity and the run time.

Must watch this video

https://youtu.be/Mo4vesaut8g

Q3. He asked me about my favorite sorting algorithm

and asked me to briefly explain the idea. Then about the complexity of quick sort.

Watch the playlist of this youtube channel then go watch visualization of each algo at a time

https://www.youtube.com/watch?v=pkkFqlG0Hds&list=PL2_aWCzGMAwKedT2KfDMB9YA5DgASZb3U

https://www.youtube.com/watch?v=kPRA0W1kECg&t=58s

Q4. From Database management he asked about he asked me questions related normalisation,

Watch the SQL, testing CI-CD from CS50 best CS course in the history of Television.

https://www.youtube.com/watch?v=Nn7EX3zkGUo&list=PLhQjrBD2T380xvFSUmToMMzERZ3qB5Ueu
then you have to go to read the stuff there and memorize some normal forms. And tell him the basic one NF1 for example in ur answer.

https://en.wikipedia.org/wiki/Database_normalization

Q5. What are Testing methods(blackbox, white box)?

Memorize this

https://www.guru99.com/back-box-vs-white-box-testing.html

Q6. Thread and processes and synchronization and how scheduling is done.

Threading watch this amazing guy tutorial 1 and 2 and if there is 3 this is the best guy ever

https://www.youtube.com/watch?v=olYdb0DdGtM&t=5s

processing basics

just watch the part of processing in neso academy video 16-23

https://www.youtube.com/watch?v=OrM7nZcxXZU&list=PLBlnK6fEyqRiVhbXDGLXDk_OQAeuVcp2O&index=16

synchronization of processes (after u watch the basics about synchronization)

What methods do you use to do synchronization ? this question was in the interview legit memorize some methods

https://en.wikipedia.org/wiki/Synchronization_(computer_science)

Q7. how scheduling is done and tell me some of the methods to do scheduling this question was in the interview iam not kidding yaaarrrrrr memorize some methods

https://www.quora.com/What-is-scheduling-and-synchronization

https://en.wikipedia.org/wiki/Scheduling_(computing) methods

important !!!!!

Q8. What is a life cycle model ?

Asked me about agile methods and waterfall method

Watch the videos from module 1 and 2 and 3

You can skip some but not recommended watch on x2 speed the guy is boring.

https://nptel.ac.in/courses/106105182

Q9. What is a model and what is a simulation? Give me some examples about simulation and a model ?

https://en.wikipedia.org/wiki/Simulation
https://en.wikipedia.org/wiki/Modeling_and_simulation

https://en.wikipedia.org/wiki/Conceptual_model_(computer_science)#:~:text=In%20computer%20science%2C%20a%20conceptual,ideas%

20in%20a%20problem%20domain.

Simulation models can be generally classified into one of three major types, namely, **continuous event simulation, discrete event simulation, and Monte Carlo simulation or Monte Carlo methods (MCM)**. In simple terms, MCM may be any procedure that uses randomly generated numbers to solve a problem.

https://mosimtec.com/types-of-simulation-models/

https://en.wikipedia.org/wiki/Monte_Carlo_method

In system engineering he asked about control,

Q10 .What is control? What methods do you use to control ?

watch part 1 and part 2 from from introduction to control systems Only

https://www.youtube.com/watch?v=PLgxjVQ6sZ8&list=PLyqSpQzTE6M8-wda5vbgHkMQTmu-21hRK&index=3

PID control is an example method of controlling

Q11. What is feedback loop system

watch these as well
https://www.youtube.com/watch?v=XMfH2P2Fc6Q

https://www.youtube.com/watch?v=7aIcsFiUigQ

https://www.youtube.com/watch?v=BJFyyqpdNg4

Give me an example of feedback loop system

https://www.watlow.com/blog/posts/setting-up-a-control-loop

what things / factors u keep in mind while designing a database:

https://datahq.co.uk/knowledge-hub/blog/top-10-considerations when-choosing-a-database-management-system

https://stackoverflow.com/questions/141126/what-is-important-to keep-in-mind-when-designing-a-database

What are the design phases of a database ?

This one he will ask you and tell him all the stages.

https://www.relationaldbdesign.com/database-design/module3/dblc design-stages.php

What is UML ?

The **Unified Modeling Language** (**UML**) is a general-purpose, developmental, modeling language in the field of software engineering that is intended to provide a standard way to visualize the design of a system.[1] The creation of UML was originally motivated by the desire to standardize the disparate notational systems and approaches to software design. It was developed at Rational

Software in 1994–1995, with further development led by them through 1996.[1]

In 1997, UML was adopted as a standard by the Object Management Group (OMG), and has been managed by this organization ever since. In 2005, UML was also published by the International Organization for Standardization (ISO) as an approved ISO standard.[2]Since then the standard has been periodically revised to cover the latest revision of UML.[3]In software engineering, most practitioners do not use UML, but instead produce informal hand drawn diagrams; these diagrams, however, often include elements from UML.[4]:536

https://en.wikipedia.org/wiki/Unified_Modeling_Language#:~:text=The%20Unified%20Modeling%20Language%20(UML,the%20design%20of%20a%20system.

Other guy's interview :

Asked me about agile methods and waterfall methods Cordula will email you about areas which cover interviews. They were 5. I had an interview for around 15 minutes. He started with my motivation for applying to TU ilmenau. Then Questions were: what is quick sort algorithm and its complexity, define waterfall method and its advantages, what is the difference bw OSI network layer 4 & 2, what things / factors u keep in mind while designing a database, and what is UML Questions from -What is a system and simulation how are they different - Sorting Algorithms with running time -Normalisation of Database -Agile and Waterfall model -OSI models physical layer and control layer These are the one i remember.

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